

Forecasts and Portfolio

Guide 5 of 6

TL;DR

Forecasts is the prediction-market surface. The firm watches markets on Polymarket and Kalshi, retrieves relevant sources from its own corpus, builds a deterministic decision trace for each market, and (only with explicit per-prediction and per-bet authorization from founder-alpha) places a paper or live bet. Live trading is guarded by eight successive human gates. After resolution, every prediction is Brier-scored, log-loss-scored, and added to the public calibration manifest. This guide explains paper vs. live, the Market Decision Trace, the eight-gate safety architecture (so you understand what's behind every public forecast), the Portfolio surface, and the resolution / override / mismatch / revision chain. Live authorization itself is operator-only and is covered in Guide 6.

You will need

- A signed-in Codex session.
- Familiarity with how conclusions work (Guide 2), because the forecast generator retrieves them as its primary inputs.

This guide does NOT cover

- Credential storage and the per-bet operator confirmation flow — operator-only, see Guide 6.
- Scheduler operations and the kill switch — Guide 6.

1 What Forecasts and Portfolio are

Forecasts (/forecasts) is a grid of prediction-market opportunities the firm has scored. Each card is one prediction. A prediction is publicly visible only after a separate authorization step that founder-alpha performs.

Portfolio (/forecasts/portfolio and the dashboard rail) is the live and paper position view: open positions, realized PnL, exchange balances, exposure relative to caps.

[screenshot screenshots/05_forecasts_grid.png not yet captured — The Forecasts grid, with one open prediction in detail view.]

2 Where the markets come from

- **Polymarket and Kalshi.** The system pulls markets from both venues on a cycle. Each market becomes a row keyed on venue and external id.
- **Operator-curated watchlist.** The firm also keeps a watchlist of specific market URLs to cover next cycle, which founder-alpha maintains.

3 How a forecast is generated

For each market the firm decides to score, the workshop:

1. Retrieves the firm's most relevant conclusions and claims via embedding search. The retrieval bundle must contain at least three distinct conclusions; otherwise the workshop abstains.
2. Checks the near-duplicate window. If a forecast on a similar market was published in the last 24 hours, the workshop abstains.
3. Checks the market-close buffer. If the market closes within one hour, the workshop refuses to predict. Stale-but-still- open markets are the firm's most expensive losses.
4. Wraps the retrieval bundle in unambiguous markers and calls a language model with a strict JSON contract: probability, confidence band, a short headline, a reasoning body, uncertainty notes, and citations.
5. Validates every quoted span against the cited source — verbatim, character-by-character. Fabrications cause the prediction to abstain.
6. Builds a **Market Decision Trace** (next section).
7. Persists the prediction with status `PUBLISHED`, a trace carrying the decision metrics, and one citation row per quoted span.

4 The Market Decision Trace

This is the artifact most worth understanding on the Forecasts side. After the language model has produced a probability and a written rationale, the workshop builds a separate object — the **Market Decision Trace** — that is *fully deterministic*: it computes a small set of metrics from the inputs, runs a fixed rule graph over them, and produces an action (`HOLD`, `WATCH`, `PAPER`, or `LIVE`) and a stake recommendation. No randomness, no further model calls.

The trace records:

- The decision metrics it computed — edge estimate (firm probability vs. market price), confidence, locality (how on-domain the retrieval was), liquidity, contradiction load, decay status.
- Each rule's firing — which threshold was crossed, which veto triggered, which combination escalated to the next tier.
- The final action and stake.
- A version string (so a later refactor that changes the trace format is detectable and not silently mixed with old data).

The deterministic decision trace is the artifact the firm is willing to defend. The model's prose explains, never overrides. If the prose says "buy" but the trace says `WATCH`, the trace wins.

[screenshot screenshots/05_decision_trace.png not yet captured — A Market Decision Trace, with the computed metrics, the rule firings, and the resulting action.]

5 Paper vs. live

The system has two trading modes:

- **Paper.** The default. The scheduler ingests markets, scores them, and writes bets to a paper-mode ledger. No order ever reaches an exchange.

- **Live.** Requires a master live-trading flag to be on plus the seven other gates below. Real funds move. Live mode is operator-only.

6 The eight-gate safety architecture

Whether or not you are the person clicking through them, it helps to know the gates exist. Each gate is a separate human action; *no gate auto-promotes the next*.

1. **Exchange credentials configured.** Polymarket and/or Kalshi credentials present on the running processes.
2. **Scheduler ingesting & monitoring.** The forecasts worker is running and writing recent rows. Absence of fresh rows blocks every downstream step.
3. **Paper mode validated.** Paper bets have been written, scored, and reconciled for long enough that the firm trusts the decision pipeline.
4. **Risk caps configured.** The maximum per-bet stake and the maximum daily loss are set to numbers the firm is willing to lose. The submitter checks both on every bet.
5. **Master live flag.** The platform-wide live-trading flag must be on. Without this, no other authorization counts.
6. **Per-prediction live authorization.** Founder-alpha sets a flag on a specific forecast row. A row that is live- eligible in principle but not yet authorized is still paper- only.
7. **Per-bet live confirmation.** For each individual bet, founder-alpha clicks confirm. The system refuses to submit without it.
8. **Kill switch clear at submit time.** The submitter re-evaluates the kill switch immediately before placing the order. If the kill switch flipped between confirmation and submission, the bet is dropped.

If any gate fails, every later gate refuses. The staircase is the control surface.

[screenshot screenshots/05_forecasts_authorization.png not yet captured — The per-bet confirmation modal: live amount, gate state, and the explicit human-confirmation click target. Visible only to founder-alpha.]

The reason for eight gates instead of one is to make it impossible to accidentally place a real-money bet through software — every gate is an explicit human decision, and the eighth gate (kill switch re-checked at submit time) is itself a fresh re-evaluation, not a cached one. The integrity of the contract is also enforced inside the schema: the database rejects any live bet that doesn't carry a matching per-prediction authorization timestamp.

7 The Portfolio surface

Portfolio (/forecasts/portfolio) shows:

- Open positions per exchange, with side, stake, entry price, and current price.
- Realized PnL by exchange and overall.
- Exposure relative to the configured caps.
- Recent fills.
- Rolling 90-day Brier score and mean log-loss.

- The kill switch state and the daily-loss cap state.

[screenshot screenshots/05_portfolio_summary.png not yet captured — The Portfolio summary card on the founder dashboard.]

The portfolio surface is read-only with respect to live trading. You cannot place a bet from the portfolio view — bets are placed from the per-forecast card by founder-alpha.

Preview — partially shipped

The portfolio's equities sub-surface (real-money equity orders) is gated identically to the prediction-market surface but currently ships in read-only mode for live accounts. The eight gates are wired in; the equity submitter is staged behind an additional flag and remains operator-only.

8 Resolution and calibration

When a market closes, the workshop polls the venue and writes a resolution carrying:

- The market outcome: YES, NO, CANCELLED, or AMBIGUOUS.
- The Brier score: $(p - y)^2$, where p is the firm's predicted probability and y is 1 (YES) or 0 (NO).
- The log-loss: $-[y \ln p + (1 - y) \ln(1 - p)]$.
- The calibration bucket (rounded predicted probability, one decimal).
- A justification and the raw settlement payload.

If a founder believes the venue resolved incorrectly (uncommon but real), the chain handles it:

- **Resolution override** — the firm records an alternative resolution with a citation and reason. Carries the actor and timestamp.
- **Resolution mismatch** — if the venue later disagrees with the override, a mismatch row is logged.
- **Resolution revision** — an append-only revision history, so prior settlement is preserved for audit.

9 The public calibration manifest

The firm publishes its own track record. `/calibration` renders a public calibration manifest — per bucket, the number of predictions and the realized YES rate. `/calibration/horizon` shows the same broken down by time-to-resolution. The manifest is regenerated on a schedule.

This is one of the most uncomfortable surfaces the firm offers: it is the firm publicly grading itself on every prediction it has staked publicly. If the firm has been consistently overconfident at the 0.7 bucket, the public page shows it.

10 The kill switch and daily loss cap

These are operator surfaces, but as a founder you should know they exist:

- **Per-submit kill switch.** Re-checked immediately before each order placement. Trips even bets that already passed operator confirmation. Open positions are not auto-closed; they are closed manually.
- **Daily-loss cap.** Once tripped, the submitter refuses new orders for the rest of the calendar day, regardless of how green the other gates are. Resets by calendar day.

11 What you actually do with Forecasts day to day

For most founders, the Forecasts surface is a reading view:

- Browse the grid to see what the firm is predicting.
- Click a card to read its decision trace and citation chain. Each citation links back to a firm conclusion, so you can verify the chain of reasoning yourself.
- Watch resolutions over time; the rolling Brier score on the portfolio card tells you whether the firm's calibration is holding up.
- Suggest watch-list additions to founder-alpha for markets you'd like the firm to score.

12 Where the docs are

- The other five guides in this series.
- Guide 6 covers the operator surfaces for Forecasts: per-bet confirmation, kill switch, resolution override, and the equities preview.

Where to read next

Continue with [Guide 6 — Operator Console](#) for what founder-alpha does behind the scenes: kill switches, live authorization, the publication review queue, the quarterly methodology review week, deletion vs. retraction. Reading Guide 6 is useful even if you don't have operator privileges — it explains the gates that everything else passes through.